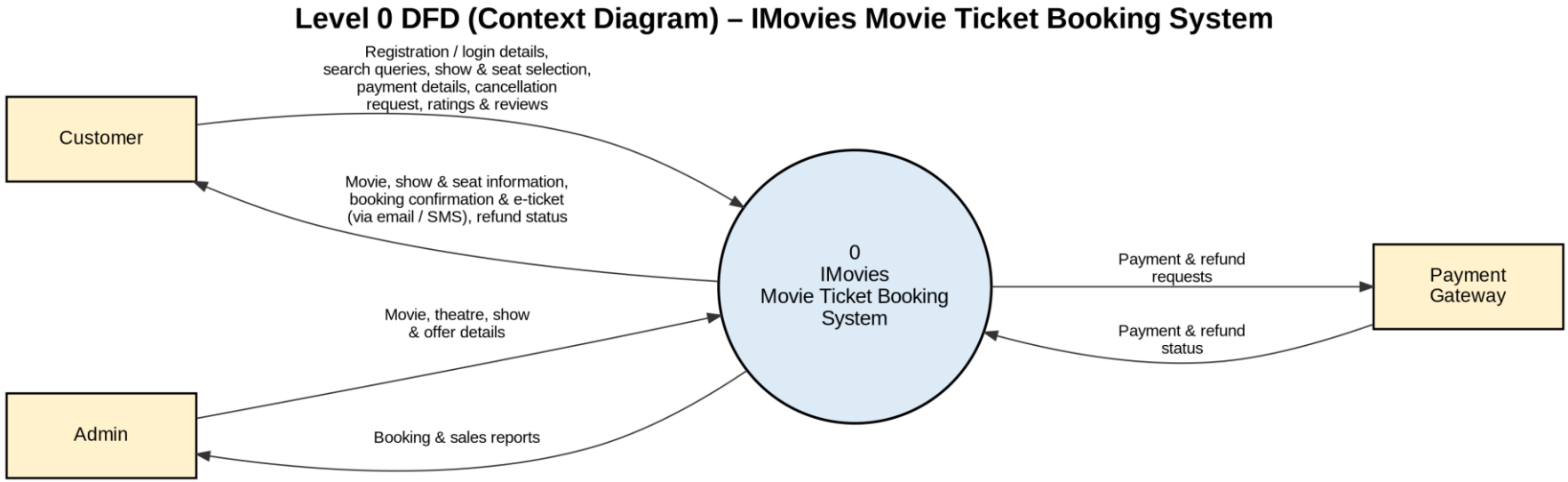
Project Design Phase-II
Data Flow Diagram & User Stories

Date	17 October 2026
Team ID	SWUID2026403471
Project Name	IMovies
Maximum Marks	4 Marks

Data Flow Diagrams:

A **Data Flow Diagram (DFD)** illustrates how data moves within the IMovies platform. It captures how users interact with the system, how information flows between different components, and where the data is stored.

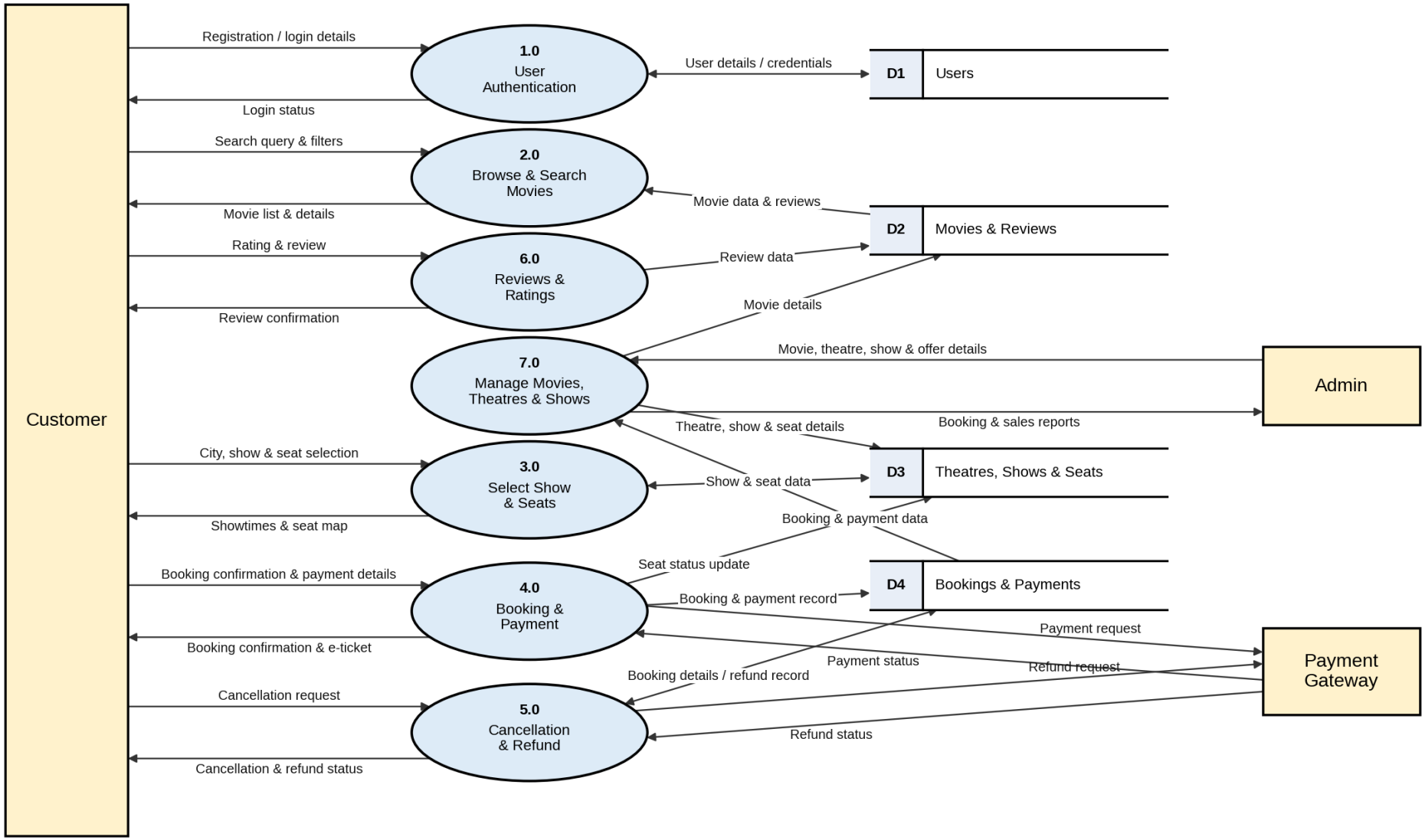
Example: DFD Level 0 (Industry Standard)



User Stories

Entity	Role and data exchanged with the system
Customer	Sends registration/login details, search queries, show and seat selection, payment details, cancellation requests, ratings and reviews. Receives movie, show and seat information, booking confirmation and e-ticket (via email/SMS), and refund status.
Admin	Sends movie, theatre, show and offer details. Receives booking and sales reports.
Payment Gateway	Receives payment and refund requests from the system. Returns payment and refund status.

Level 1 DFD – IMovies Movie Ticket Booking System



Level 1 – Process descriptions

No.	Process	Description
1.0	User Authentication	Registers new users, verifies login credentials, and allows password reset and change using the Users store.
2.0	Browse & Search Movies	Shows Now Showing and Upcoming movies, search and filters by language, genre and rating, and movie details with ratings.
3.0	Select Show & Seats	Lets the customer choose a city, theatre, date and showtime, and pick seats from the seat map using show and seat data.
4.0	Booking & Payment	Confirms the booking, applies coupons, sends the payment request to the Payment Gateway, saves booking and payment records, updates seat status and sends the e-ticket (QR code) by email/SMS.
5.0	Cancellation & Refund	Cancels a booking, sends the refund request to the Payment Gateway, updates booking and refund records and informs the customer of the status.
6.0	Reviews & Ratings	Accepts ratings and reviews from customers and stores them so they can be shown with movie details.
7.0	Manage Movies, Theatres & Shows	Lets the Admin add and update movies, theatres, shows and offers, and view booking and sales reports.

Level 1 – Data stores

ID	Data store	Contents
D1	Users	User profiles and login credentials.
D2	Movies & Reviews	Movie details (title, language, genre, cast, trailer) and customer ratings and reviews.

ID	Data store	Contents
D3	Theatres, Shows & Seats	Theatres, screens, show timings, seat layout and seat status.
D4	Bookings & Payments	Booking records, payment and refund records, and e-ticket details.